

MATHESH S

Aspiring Design Engineer | CAD Modeling · FEA · CFD · Kinematic Simulation

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PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate with a hands-on, simulation-driven approach to product design, seeking a Design Engineer role (internship / remote / entry-level) in the automotive, aerospace, or industrial sectors. Proficient across the full design-validation pipeline — from parametric 3D modeling in **SolidWorks**, **Solid Edge**, and **Fusion 360** to structural and fluid analysis in **ANSYS (Static Structural & Fluent)**. Demonstrated ability to translate theoretical physics — stress theory, fluid dynamics, kinematics — into functional, manufacturable designs validated against classical engineering theory and empirical (NASA-benchmarked) data. Passionate about engineering systems where geometry, motion, and load interact, and committed to closing the gap between CAD concept and real-world performance.

CORE COMPETENCIES

CAD / 3D Modeling	Simulation & Analysis	Supporting Skills
SolidWorks (Parametric Modeling, Mating, Tolerancing)	ANSYS Fluent (CFD)	Engineering Tolerancing & Interference Checks
Solid Edge (Structural & Geometric Constraints)	ANSYS Static Structural (FEA)	Kinematic & Dynamic Motion Analysis
Fusion 360 (Advanced Surface Modeling)	Finite Element Analysis (FEA) Computational Fluid Dynamics (CFD)	Embedded Systems & Relay-Based Automation Web-Based 3D Visualization (HTML, CSS)

PROJECT EXPERIENCE

4-Cylinder Engine Assembly — Internal Combustion Engine Design

SolidWorks | Parametric Modeling · Dynamic Motion Analysis

- Designed a full parametric 3D assembly of a 4-cylinder internal combustion engine, modeling all major components — pistons, crankshaft, connecting rods, cylinder block, and valvetrain — with precise mating relationships.
- Engineered strict tolerance stacks and ran interference checks across mating components to ensure assembly feasibility and mechanical fit.
- Simulated dynamic motion of the piston-crankshaft mechanism to validate kinematic behavior and structural response under operating loads.

Car Suspension & Steering System — Structural & Kinematic Design

Solid Edge | Load Distribution · Joint Articulation

- Modeled a complete 3D suspension and steering assembly, defining geometric constraints for control arms, tie rods, and steering linkages.
- Analyzed load distribution across chassis mounting points to ensure structural integrity under simulated driving loads.
- Evaluated joint articulation, spring compression behavior, and linkage travel limits to validate suspension geometry through its full range of motion.

Turbulent Flow Over a Backward-Facing Step — NASA-Validated CFD Study

ANSYS Fluent | Turbulent CFD · Empirical Validation

- Conducted a computational fluid dynamics simulation of turbulent flow separation and reattachment over a backward-facing step.
- Validated simulated wall shear stress distribution and flow reattachment length against published NASA empirical benchmark data, confirming computational accuracy of the turbulence model.

Cantilever Beam FEA — Static Structural Analysis

ANSYS Static Structural | FEA · Stress Analysis

- Performed a static structural finite element analysis on a cantilever beam under end-loading conditions.
- Evaluated maximum total deformation (0.1925 mm) and equivalent (von Mises) stress distribution, cross-validating FEA results against classical Euler-Bernoulli beam theory for accuracy.

Offline Voice-Controlled Switch — Hardware & AI Module Integration

Ai-Thinker VC-02 (Offline Speech Recognition) | Electronics Integration · Automation

- Designed and built a localized voice-controlled switch using the Ai-Thinker VC-02 offline speech recognition module, eliminating dependency on cloud-based voice processing.
- Integrated a 4-channel relay module to independently actuate multiple electrical components via voice command, without requiring internet connectivity.

EDUCATION

Bachelor of Engineering (B.E.), Mechanical Engineering

Sri Krishna College of Engineering and Technology (SKCET), Coimbatore | August 2025 – Present

ADDITIONAL SKILLS & INTERESTS

- Interactive 3D model embedding for web-based portfolio visualization (HTML, CSS)
- Strong grounding in engineering mechanics, thermodynamics, and fluid dynamics fundamentals underpinning simulation work
- Detail-oriented approach to design-for-manufacturability and engineering tolerancing
- Interests: Automotive powertrain design · Aerospace structural analysis · Computational simulation of physical systems